

Rail-to-rail low-power high-slew-rate CMOS analogue buffer

J.M. Carrillo, R.G. Carvajal, A. Torralba and J.F. Duque-Carrillo

A low-power rail-to-rail CMOS analogue buffer is presented. The circuit is based on an input stage made up of two complementary class AB differential pairs, while a simple additional circuit allows rail-to-rail operation at the output terminal. The proposed circuit combines low static power consumption and high drive capability, resulting in suitability for applications with large capacitive loads. Simulated results are provided.

Introduction: Analogue voltage buffers are important building blocks in mixed signal designs, where they are used for signal monitoring or for driving loads [1–3]. In the first case, the buffer is usually connected to an internal node of the circuit under test and a low input capacitance is required, as any increase of the parasitic capacitance at this node could be critical. However, when the buffer is used to drive loads a high slew-rate along with a wide output signal swing is desirable in order to drive the load as fast as possible over the entire voltage range.

The supply voltage of present integrated circuits has been reduced, attending to power consumption and reliability issues. This trend has forced most analogue basic building blocks to be redesigned, in an attempt to guarantee their overall performance. Under these design constraints, rail-to-rail operation has become mandatory in low-voltage designs, in order to maximise the signal-to-noise ratio.

In this Letter we introduce a circuit technique to obtain a rail-to-rail CMOS analogue buffer with class AB behaviour, leading to an approach with low power consumption and high drive capability.

Previous implementation: Fig. 1a illustrates a p-channel class AB differential pair, able to deliver a large amount of current when a large differential signal is applied to its input terminals. The impedance at node A in the differential pair is very low and its voltage is approximately constant even in the case of large input signals. Consequently, a differential voltage $V_1 - V_2$ leads to large current variations in M2 and, consequently, in M3, as shown in Fig. 1b.

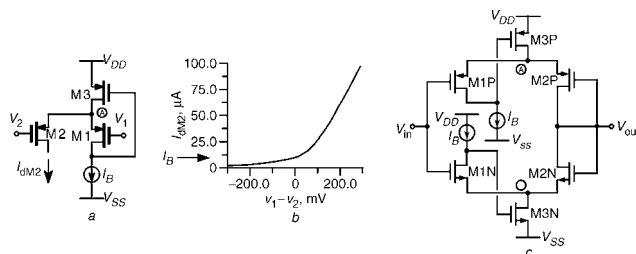


Fig. 1 Class AB differential input cell; its DC transfer characteristic for a differential input signal; and low power buffer based on two complementary class AB differential input cells

a Class AB differential input cell
b DC transfer characteristic
c Low power buffer

A class AB voltage buffer can be obtained by combining two complementary differential cells [3], as illustrated in Fig. 1c. Nevertheless, the circuit in Fig. 1c has two main limitations. First, the gate-source voltage of M3P and M3N may force the driver transistors M1P and M1N, respectively, to operate in the triode region, reducing the available operating voltage range. This drawback can be overcome by introducing voltage level shifters to drive M3P and M3N, as will be explained later. Secondly, the output voltage swing of this structure is limited when the output node goes close either to the positive or the negative supply rail, due to the constraints of the p-channel and n-channel input differential pairs to operate close to V_{DD} and V_{SS} , respectively.

Proposed analogue buffer: Fig. 2 illustrates the transistor-level implementation of the proposed rail-to-rail CMOS analogue buffer. The circuit is a single-gain-stage in which the input branch is made up

of two complementary class AB differential pairs. One important difference with respect to the circuit in Fig. 1c is that in this case the output node is not driven directly by the input drivers, but current mirrors M4P–M5P and M4N–M5N, respectively, are used with this goal. As a consequence, the common gate of M2P and M2N is now the non-inverting input terminal.

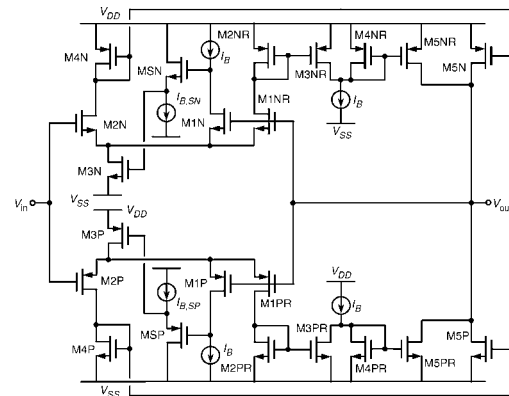


Fig. 2 Proposed rail-to-rail class AB buffer

In the midsupply region both the PMOS and the NMOS input pairs are active and their biasing currents are mirrored to the output branch of the circuit through current mirrors M4P–M5P and M4N–M5N, respectively. This configuration allows the NMOS input pair driving the output node in the voltage region close to V_{DD} , whereas the PMOS pair controls the output terminal in the voltage range close to V_{SS} . Unfortunately, close to V_{DD} the p-channel input pair is cut off and no current is mirrored to the bottom part of the output branch, turning off the buffer. A similar situation arises close to V_{SS} , where the n-channel differential pair is not active. For this reason, transistors M1PR–M5PR and M1NR–M5NR have been included in the circuit in Fig. 2, maintaining active the output branch over the entire voltage range.

Thus, the operation of the buffer can be detailed as follows. When the input signal, V_{in} , is in the midsupply region, the two input pairs, M1P–M2P and M1N–M2N, are active and M4P–M5P and M4N–M5N mirror a current equal to I_B to the output branch. Furthermore, a replica of the current I_B is copied through transistors M1PR (M1NR) and M2PR–M3PR (M2NR–M3NR), providing the current required by the current source in the additional circuit at the bottom (top) of the output branch. As a consequence, transistors M4PR and M5PR (M4NR and M5NR) are turned off and do not contribute to the output current. When the input signal goes close to V_{DD} , the PMOS input pair cuts off and the replica of the inverting input branch, i.e. M1PR–M3PR, does not send any current contribution to the current source in the additional circuit at the output. Thus, M4PR and M5PR turn on, drawing a current equal to I_B from the output branch and, thus, maintaining the buffer turned on. Something similar happens when the input signal, V_{in} , is close to V_{SS} .

It should be noted that voltage level shifters have been included in the input stage so as to drive transistors M3P and M3N, avoiding the operation of M1P and M1N, respectively, in the linear region and extending the input branch signal range up to both supplies. Therefore, rail-to-rail operation is reached at the input as well as at the output terminal of the circuit.

The dynamic operation of the proposed buffer is enhanced by the high drive capability of the class AB differential pairs in the input branch of the circuit. In the case of a large input signal in the positive direction, transistor M2P cuts off whereas transistor M2N draws a large amount of current, which is mirrored to the output branch by M4N and M5N. Conversely, when a large input signal is applied in the negative direction, transistor M2N cuts off and M2P delivers a large current, which is copied to the output branch by M4P and M5P.

The input capacitance of the proposed buffer can be reduced by scaling down the size of transistors M2P and M2N with respect to the geometry of M1P and M1N. Nevertheless, it must be pointed out that the reduction of the aspect ratio of these transistors would lead to a decrease of their effective drive capability. Besides, the bandwidth can be very large as there is only one high impedance node in the circuit. However, the single-gain-stage configuration along with the high output impedance of the output node makes this approach more suitable to be

used for driving large capacitive loads, given that low resistive loads would reduce the total gain of the buffer and, consequently, its accuracy.

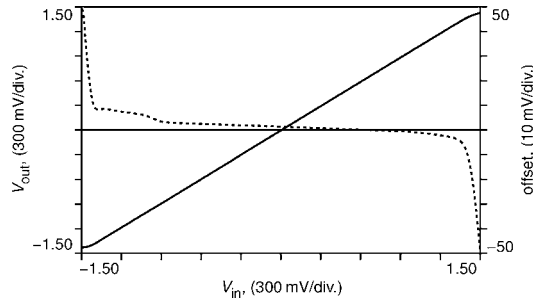


Fig. 3 DC transfer characteristic of analogue buffer in Fig. 2

— output voltage - - - - - offset voltage

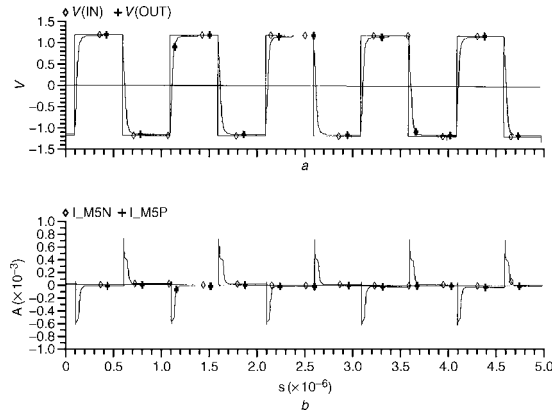


Fig. 4 Large-signal transient response of buffer in Fig. 2 for $2.4 V_{pp}$, 1 MHz square input signal with 10 pF load

a Input and output voltages b Currents through output transistors

Simulated results: The analogue voltage buffer in Fig. 2 was designed in a 0.35 μm CMOS technology to operate with a supply voltage of ± 1.5 V, a biasing current of 10 μA , and a capacitive load of 10 pF.

Fig. 3 shows the DC transfer characteristic of the proposed analogue buffer along with the offset voltage. As expected, rail-to-rail behaviour is achieved. Fig. 4 illustrates the large-signal transient response of the circuit in Fig. 2 to a $2.4 V_{pp}$ 1 MHz input square signal. Particularly, the

output voltage reveals a high slew-rate due to the class AB operation of the input stage. However, the large ratio between the maximum current and the quiescent biasing current flowing through the output transistors demonstrates that the proposed solution leads to low power consumption and high drive capability.

The open-loop gain and unity-gain frequency, simulated for a DC input voltage equal to zero, were approximately 54 dB and 6.1 MHz. The relatively low value of the gain is due to the single-gain-stage arrangement of the circuit. The value of the GBW can be increased at the expense of increasing the biasing current of the input differential pairs and, so, raising the power consumption. A THD equal to -44.6 dB is obtained for a $2.4 V_{pp}$ 100 kHz input sinewave signal and the simulated input capacitance of the proposed buffer is lower than 32 fF when no scaling is performed in the input transistors.

Conclusion: A rail-to-rail voltage buffer with reduced input capacitance is proposed. Rail-to-rail operation is achieved at the input as well as the output terminal of the circuit. The class AB behaviour of the introduced circuit leads to a low power consumption and a high slew-rate, resulting in it being very suitable to drive large capacitive loads. Simulated results have been provided to show the operation of the circuit.

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